

Trainer Debrief — Lab 11

Continuous Gage R&R — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Validate a continuous measurement system for your capstone using Gage R&R (A4).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-11-gage-rr-continuous.xlsx

%Study Variation for Total Gage R&R lands around 38-42% (>30% = UNACCEPTABLE) and the number of distinct categories is 3 (<5 = inadequate). Reproducibility is the larger share: N. Faridah reads ~5.9 kPa high relative to T. Chandra. Fix the operator method (a written test procedure and re-training) before re-running the study; do NOT analyse baseline capability on this measurement system.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your study uses parts spanning the full process range with randomised blinded order, and you have computed %study variation and ndc with a stated accept/reject decision.

Where this sits in the capstone

A full re-derivation of measurement system analysis for continuous data. Before any of the statistics in Days 3 and 4 can be trusted, the measurement system must be proven — an invalid gauge invalidates every conclusion that follows. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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